

## Spring 2019 – MATH 4661/6661 Analysis I – Course Syllabus

**Instructor:** Prof. Xiaojing Ye ([xye@gsu.edu](mailto:xye@gsu.edu)), <https://math.gsu.edu/xye>, Office: 25 Park Place, 1436

**Office Hours:** MW 4:30—5:30 PM or by appointment.

**Class Time and Location:** MR 7:15—8:30 PM, Langdale Hall 201.

### Reference Textbooks

- Principles of Mathematical Analysis (Third Edition) by W. Rudin. Published by McGraw-Hill
- Real Analysis and Foundations (Third Edition) by G. Krantz. Published by Chapman and Hall/CRC

### Prerequisites

Grade C or higher in Introductory Linear Algebra (MATH 3435). Co-requisite: Linear Algebra (Math 4435).

### Course Outline

Chapter 1 The real and complex number systems (6 lectures)

Chapter 2 Basic topology (6 lectures)

Chapter 3 Numerical sequences and series (9 lectures)

Chapter 4 Continuity (6 lectures)

### Course Description

Rigorous approach to calculus; proof techniques in the area of real analysis.

### Homework

There are 8 homework (5 points each) during the semester. Homework assignments are due at the beginning of the lectures on the due days. **Late homework will not be accepted.**

### Tests

There are 3 tests. The tests (20 points each) are scheduled in class on **February 13th, March 13th, and April 22nd**. Closed book/notes in all tests. There will be **no make-up tests except in an extreme verifiable emergency**. Absence from any test will result in a grade of F for the course unless arrangements are made **prior** to its administration.

### Grading Scale

Homework (40pts). Tests 1 (20pts), 2 (20pts), and 3 (20pts). Total is 100 points.

### Grading Scale

A+ (97-100); A (93-96); A- (90-92); B+ (87-89); B (83-86); B- (80-82); C+ (77-79); C (70-76); D (60-69); F (<60).

### Academic Honesty

All work submitted by a student for a grade must have been done without help from other persons. **Cheating and plagiarism are strictly forbidden.** A first occurrence will result in a grade of 0 on the assignment for **all concerned parties** as well as an Academic Dishonesty form being filed with the Dean of Students. A second occurrence will result in a grade of F for the course for the concerned parties and a second Academic Dishonesty form being filed. See the University's policy on academic honesty at <http://codeofconduct.gsu.edu/>.

### Remarks

- This course syllabus provides a general plan for the course; deviations may be necessary.
- Your constructive assessment of this course plays an indispensable role in shaping education at Georgia State. Upon completing the course, please take time to fill out the online course evaluation at the end of the semester.